

CSCI3240 Introduction to Computer Vision

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Syllabus - weekly schedule (15 weeks)

Week 01: Course overview, images as arrays, color spaces

Week 02: Point operations, histograms, contrast and gamma

Week 03: Linear filtering, convolution, Gaussian and box filters

Week 04: Edge detection: Sobel, Canny, gradients

Week 05: Image pyramids, scale space, resampling

Week 06: Feature detection: Harris corners, blobs

Week 07: Feature description and matching: SIFT, ORB

Week 08: Midterm; geometric transforms, homographies

Week 09: Image stitching, RANSAC

Week 10: Camera model, calibration, stereo basics

Week 11: Segmentation: thresholding, k-means, graph cuts

Week 12: Optical flow and tracking

Week 13: Convolutional networks for classification

Week 14: Object detection: YOLO-style detectors

Week 15: Project presentations and final review

Labs (Fridays, room 412):

Lab 01: Image basics with numpy

Lab 02: Filtering and edges

Lab 03: Features and matching

Lab 04: Homography and stitching

Lab 05: A small CNN classifier

Grading: labs 30%, midterm 30%, final project 40%.